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$$y(x) := \sqrt{x}$$

$$clz(x) := \begin{cases} 32 & \text{if } x = 0 \\ 32 - \operatorname{trunc}\left(\log_2\left(\operatorname{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right) & \text{else} \end{cases}$$

$$msb(x) := 32 - clz(x)$$

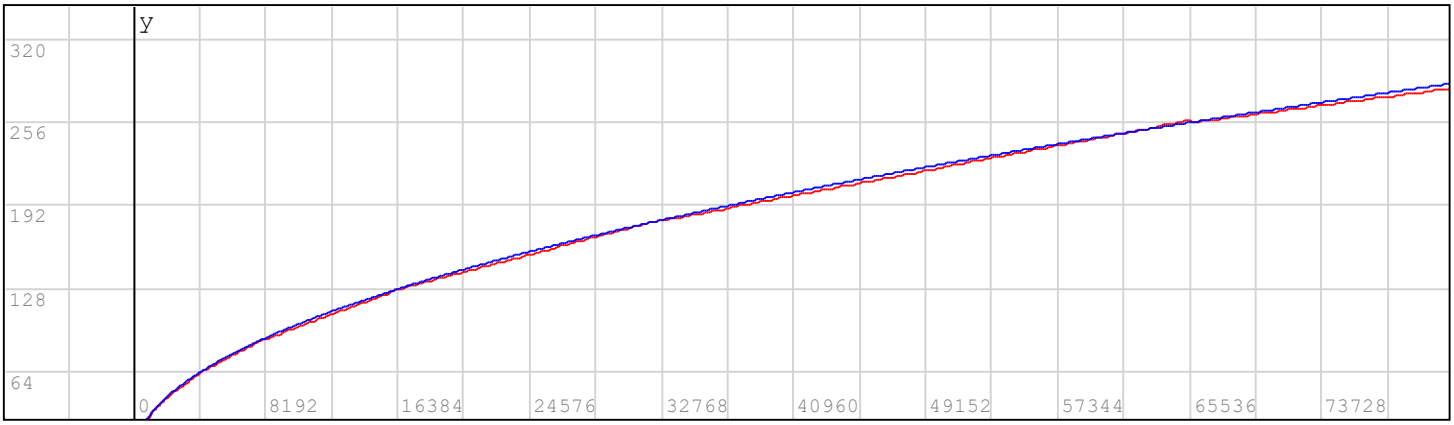
$$linScale(x) := \operatorname{trunc}\left(\frac{\operatorname{trunc}\left(\left(\sqrt{2}\right)^{msb(x)}\right) \cdot \operatorname{trunc}\left(\left(1+\sqrt{2}\right)\cdot 2^{29}\right)}{2^{32}}\right)$$

$$linSqrt(x) := \operatorname{trunc}\left(\left(\frac{\left(\operatorname{trunc}(x) - 2^{msb(x)-1}\right)}{2^{msb(x)-1}}\right) \cdot linScale(x)\right) + \operatorname{trunc}\left(\sqrt{2^{msb(x)-1}}\right)$$

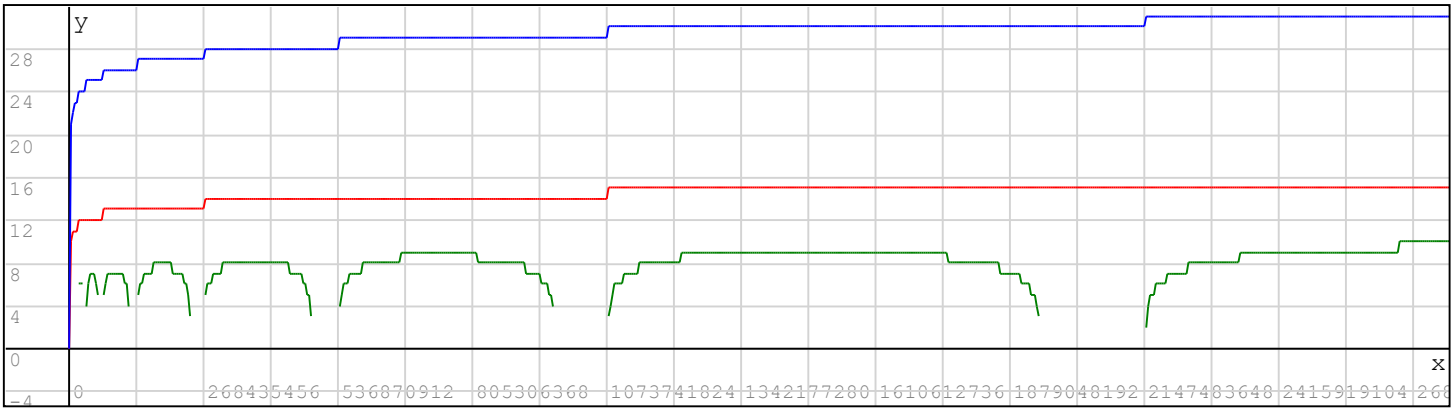
$$argBitCount(x) := \operatorname{trunc}\left(\log_2\left(x + \frac{2^{32}-1}{2^{32}}\right)\right)$$

$$resultBitCount(x) := \operatorname{trunc}\left(\log_2\left(\sqrt{\operatorname{trunc}(x)} + \frac{2^{32}-1}{2^{32}}\right)\right)$$

$$linearBitError(x) := \operatorname{trunc}\left(\log_2\left(\left(\sqrt{\operatorname{trunc}(x)} - linSqrt(x)\right)\right) + \frac{2^{32}-1}{2^{32}}\right)$$



$$\begin{cases} \sqrt{x} \\ linSqrt(x) \end{cases}$$



$$\begin{cases} argBitCount(x) \\ resultBitCount(x) \\ linearBitError(x) \end{cases}$$

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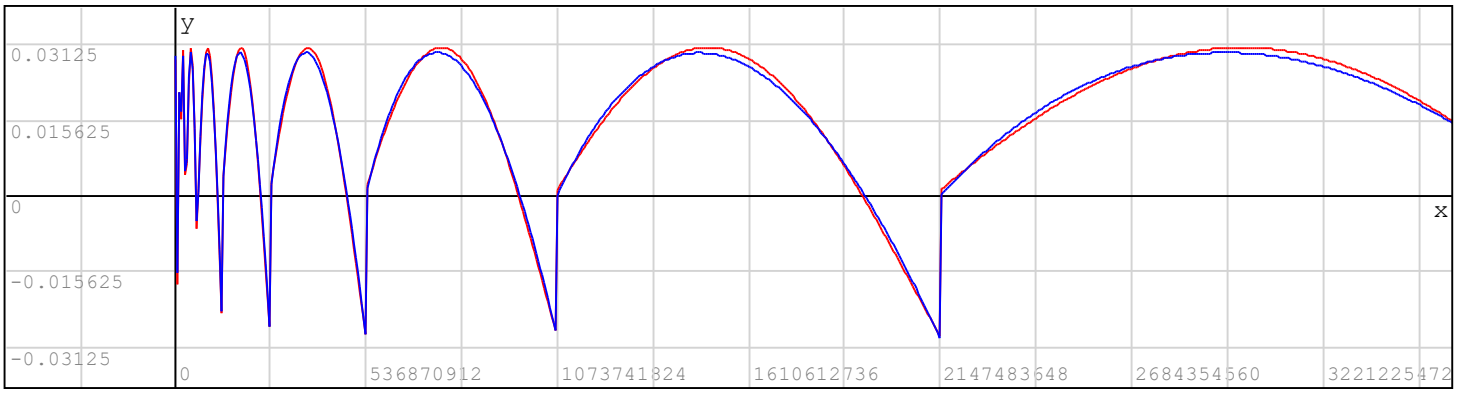
$$linSqrtErr(x) := \sqrt{x} - linSqrt(x)$$

$$sinShift := \frac{\sqrt{2}}{5} + 0.105 \qquad sinPhaseScale := 1 - 0.074 \qquad sinAngle := \frac{24.3}{360} \qquad sinScale := 0.965$$

$$PhaseK := \mathrm{trunc}\left(sinPhaseScale \cdot 2^{32}\right) \qquad PhaseOffs := \mathrm{trunc}\left(sinAngle \cdot 2^{32}\right) \qquad Scale := \mathrm{trunc}\left(sinScale \cdot \frac{2^{32}}{2^{6}}\right)$$

$$AmpShift := \mathrm{trunc}\left(sinShift \cdot \left(sinScale \cdot \frac{2^{32}}{2^{6}} \cdot \frac{2^{3}}{\left(1+\sqrt{2}\right)}\right)\right)$$

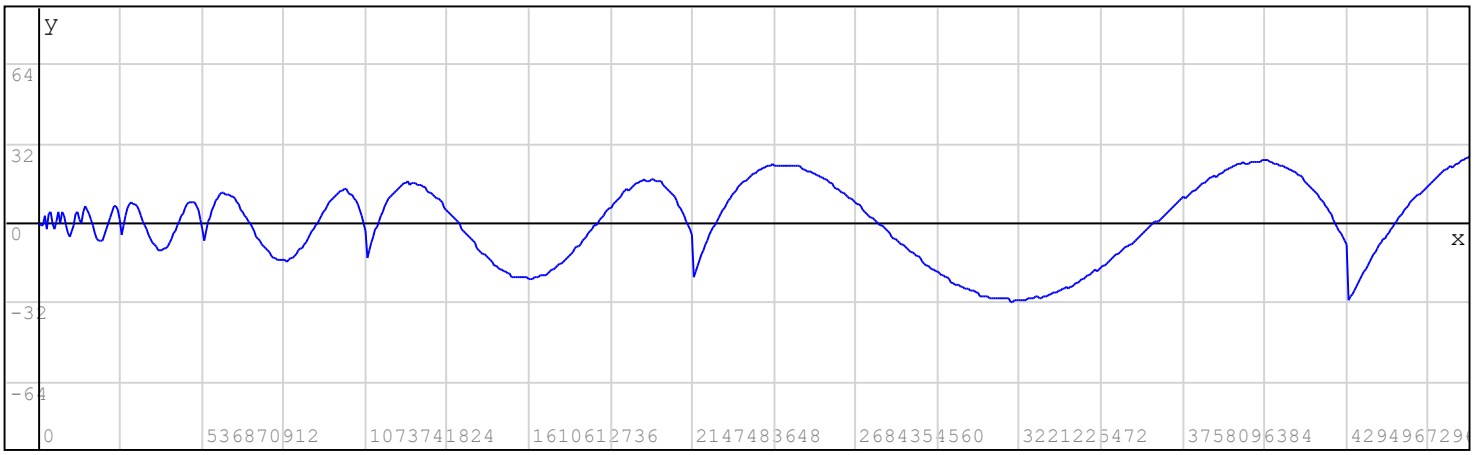
$$ec(x) := \left(\frac{\mathrm{trunc}\left(\sin\left(\frac{\mathrm{trunc}\left(\left(\frac{x - 2^{msb(x)-1}}{2^{msb(x)}} \cdot \frac{PhaseK}{2^{32}} + \frac{PhaseOffs}{2^{32}}\right) \cdot 2^{32}\right)}{2^{32}} \cdot 2 \cdot \pi\right) \cdot \left(2^{32}-1\right) \cdot \frac{Scale}{2^{32}}\right)}{2^{32}} - \frac{AmpShift}{2^{32}}\right)$$



$$\begin{cases} \dfrac{linSqrtErr(x)}{linScale(x)} \\ ec(x) \end{cases}$$

$$cosSqrt(x) := linSqrt(x) + \left(ec(x) \cdot linScale(x)\right)$$

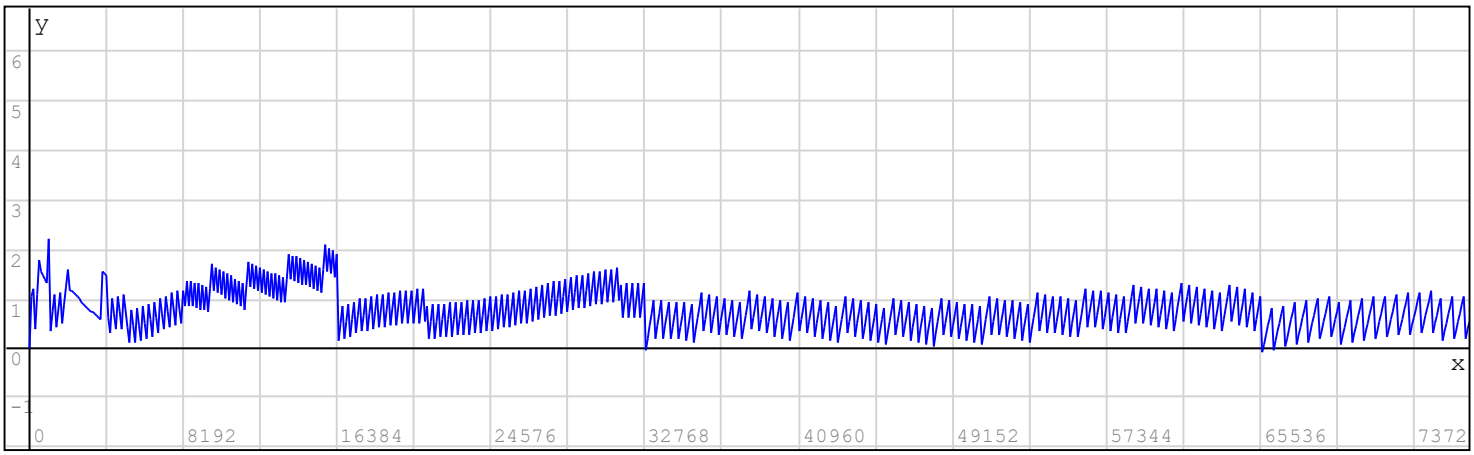
$$cosBitError(x) := \mathrm{trunc}\left(\log_{2}\left(\left(\sqrt{\mathrm{trunc}(x)} - cosSqrt(x)\right)\right) + \frac{2^{32}-1}{2^{32}}\right)$$



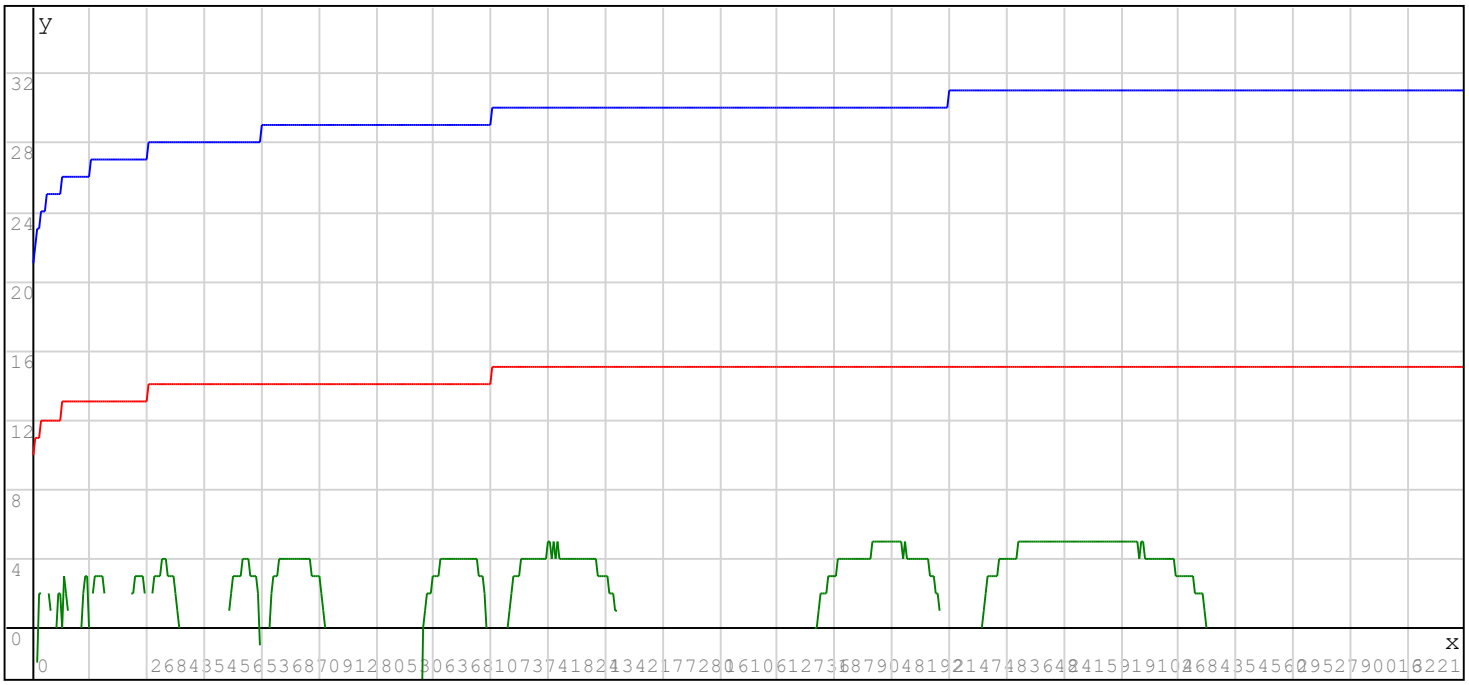
$$\left\{\sqrt{x} - cosSqrt(x)\right\}$$

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$\{\sqrt{x} - cosSqrt(x)$



$\{argBitCount(x)$
$resultBitCount(x)$
$cosBitError(x)$

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